E-Commerce Sales Interactive Dashboard Project Report

Project Name **:** Interactive E-Commerce Sales Dashboard Using Excel

Tools Used **:** Microsoft Excel (Pivot Tables, Data Cleaning, Calculated Columns Charts)

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Datasets **:** https://www.kaggle.com/datasets/faresashraf1001/supermarket-sales

I. Project Overview

This project is to analyze e-commerce sales performance by transforming raw sales data (supermarket dataset) into a clean, interactive, and visual dashboard. It allows users to monitor the monthly revenue trend, region and category performance, top products, and customer segment breakout.

II. Data Cleaning & Preprocessing Challenges

The raw dataset contains inconsistent date formats, with mixed **dd/mm/yyyy** and **mm/dd/yyyy** values. It causes incorrect year and month grouping, wrong pivot table results, and inaccurate trend analysis. Manual changes into the incorrect format is needed to make better performance.

III. Dashboard Structure & Design Decision

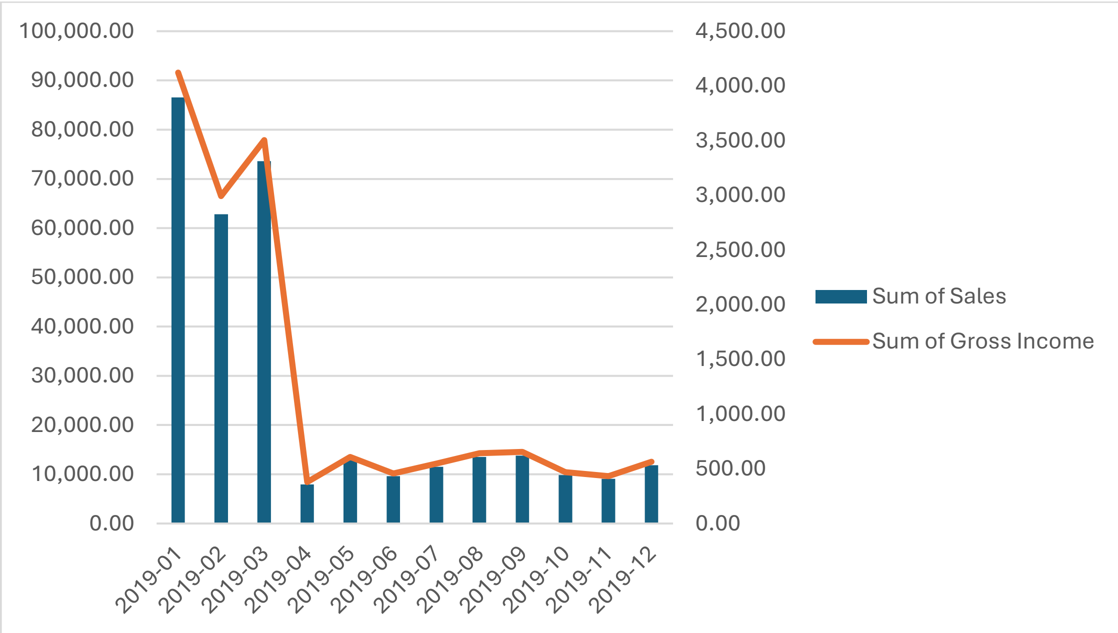
The final dashboard consists of trend chart, segment analysis, and product performance visuals. Several custom adjustments were made to improve readability and professionalism.

1. **Combo Chart (Column + Line)** replaces dual line charts to eliminate overlapping sales and profit lines. Sales is represented as columns (primary axis) and profit as a line (secondary axis) for clear trend comparison.
2. **Date Axis formatting** fixed large visual gaps in monthly trends by using real date values instead of text-based months.
3. **Donut Chart** shows sales proportion between only two customer groups: Member and Normal.

IV. Key Findings & Data Insights

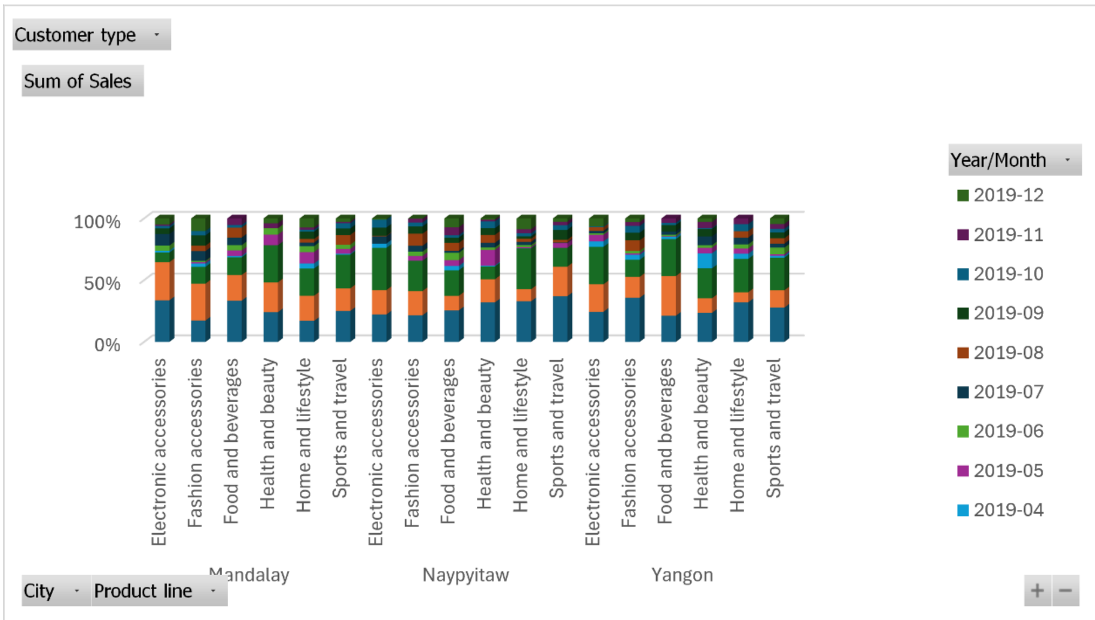
1. Monthly revenue

From the chart, it shows that there is a huge gap in sales and gross income (profit) at 2019-03 and 2019-04. During these two months, the business recorded extremely high sales volume and total revenue. This indicates that the business achieved top-line sales growth through low-margin sales activities in these months, such as heavy promotional discounts, low-profit product sales, or increased operational and procurement costs. It proves that high sales revenue does not equate to high business profitability, and blind sales expansion without cost control will severely compress profit margins.



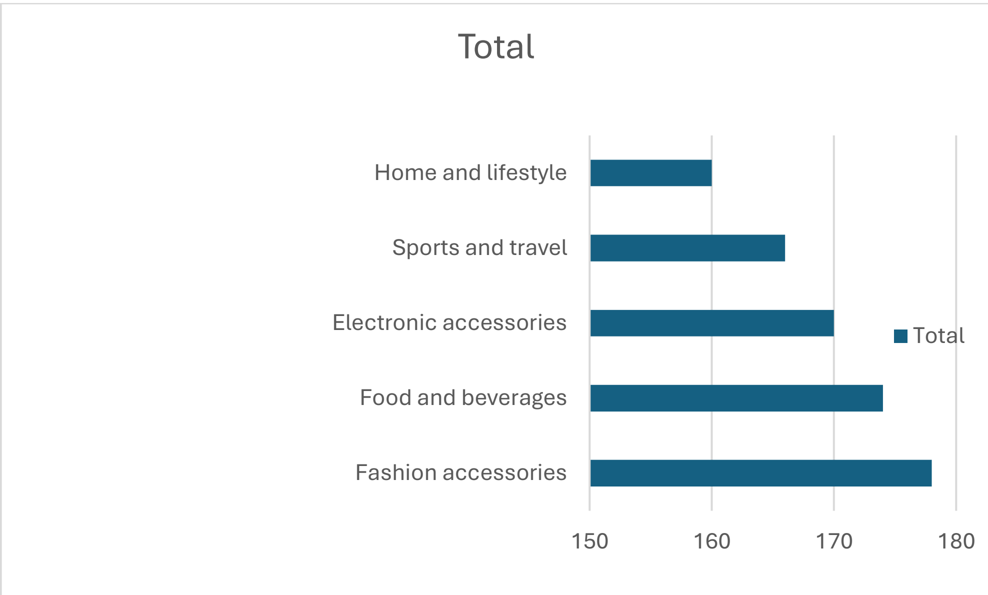
2. Regional and Category Performance

The stacked column chart shows the comparison of monthly sales proportion for each product category across Mandalay, Naypyitaw and Yangon. Sep-Dec 2019 contributes a large share of total sales for many product lines across all cities. However, monthly sales composition for the same product category varies by city, showing regional differences in seasonal buying behaviour.



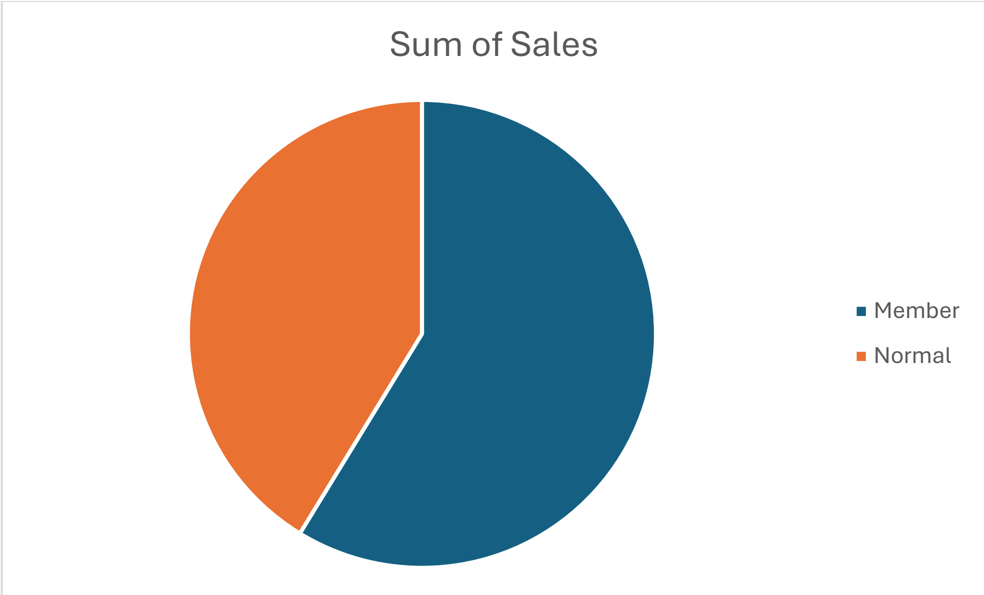
3. **Top Five Products**

The chart shows that Fashion accessories has the highest sales with the count of sales of 178.



4. **Member vs Normal Customer Behavior**

The table and chart show that member customers provide more stable and higher value between members and normal customers. It proves that customer loyalty programs positively impact sales performance.



V. Business Recommendations

Suggests company about promotional timing should be customised per city-product combination instead of applying identical campaigns to all regions.

VI. Project Learning & Conclusion

This project demonstrates the ability to clean messy real-world data, solve date inconsistency problems, build dynamic pivot table analysis, and design a fully interactive business dashboard. The dashboard successfully transforms raw data into actionable business insights regarding sales trends, regional performance, product profitability, and customer segment behaviour. The skills applied include data cleaning, formula creation, pivot table analysis, visual chart optimization, and business data storytelling — essential skills for data analysis and business intelligence roles.